

Session 2 (FIND & READ): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · #ClearDataClearImpact · Week 2 · Saturday

Name / group: _____ Date: _____

> How to use this sheet.

- > - Recap quiz on Session 1: on your own (8 minutes, start of the session).
- > - Exercise 1 (Boolean search builder): **on your own**, segment 2.1 (5 minutes).
- > - Exercise 2 (paraphrasing): at home.
- > - Exercise 3 (reading Paper 1's abstract): **on your own**, segment 2.3 (12 minutes).
- > - Exercise 4: your **group's Proposal builder 2** (33 minutes).
- >
- > The answers are discussed in the room. There is no mark.

Recap quiz: Session 1 (*individual, 8 minutes*)

1. Checking whether fluid balance is charted daily as the protocol says: research, audit or QI?

2. "In ICU staff, does hand-hygiene feedback vs none increase compliance at 3 months?" P
_____ I _____ C _____ O _____ T _____
3. Which FINER letter did our running example fail? _____
4. Which is an objective: "to understand delirium" or "to estimate the proportion with delirium"?

5. Write H0 for: "Does early mobilisation shorten ICU stay?"

The running example (for reference)

P: Population	Adults (18+) with septic shock in the ICU
I: Intervention	Restrictive fluid strategy
C: Comparison	Usual care (standard fluids)
O: Outcome	Death from any cause
T: Time	Within 28 days

Question: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*

Exercise 1: Boolean search builder (*individual, 5 minutes, segment 2.1*)

Rules: join synonyms inside a block with **OR**. Join blocks with **AND**. Put each block in **brackets**. Write operators in CAPITALS. Use English search words.

Warm-up: which gives MORE hits? (circle)

- septic shock AND fluids **OR** septic shock OR fluids
- sepsis **OR** sepsis NOT neonatal

Build your search from your group's PICO question (Session 1) and your synonym list. The comparison (C) is usually *not* searched.

Block	Concept	Synonyms (2–4). Mark possible MeSH terms with (M)
P		
I / E		
O		

My search string:

(_____ OR _____ OR _____) AND (_____ OR _____ OR _____) AND (_____ OR _____)

One filter I would add, and why: _____

Swap with your neighbour. One synonym they suggest adding: _____

The search string from the live demo (copy it here to run at home):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*" [tiab] OR "restrictive fluid*" [tiab] OR "conservative :
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Field tag / symbol	Meaning
[Mesh] or [mh]	MeSH term (includes narrower terms automatically)
[tiab]	word in the title or abstract
[pt]	publication type, e.g. randomized controlled trial[pt]
"..."	exact phrase
*	any ending (truncation): restrict* = restrict, restricted, restriction, restrictive

Exercise 2: Citing vs copying (paraphrase safely) (*individual, at home, 10 minutes*)

Method: read → close the paper → write from memory → check the facts → cite.

Patchwriting (copying the sentence and swapping a few words) is plagiarism **even with a reference**.

Part A: Spot the problem. The source is the Results section of Paper 2's abstract (Hyun 2023):

> "The positive fluid groups in both the D2 (HR, 2.13; 95% CI 1.48–3.06; P < 0.001) and D3 (HR, 1.56; 95% CI 1.10–2.22; P = 0.012) cohorts had significantly higher mortality rates than the negative

fluid groups."

For each version, tick **OK** or **Problem**, and say why.

Version	OK	Problem	Why?
a) "The positive fluid groups in the D2 and D3 cohorts had significantly higher death rates than the negative fluid groups [2]."	☐	☐	
b) "In a matched registry cohort, a positive cumulative fluid balance on ICU days 2 and 3 was associated with higher 28-day mortality (hazard ratios 2.13 and 1.56) [2]."	☐	☐	
c) "Giving more fluid on days 2 and 3 doubles the risk of death [2]."	☐	☐	
d) "In a matched registry cohort, a positive cumulative fluid balance on ICU days 2 and 3 was associated with higher 28-day mortality." (no citation)	☐	☐	

Part B: Your turn. Paraphrase this sentence from Paper 1's abstract in your own words, with a citation:

> "After 28 days, mortality and the numbers of days alive without life support were similar between groups."

My paraphrase: _____

Check: different structure? ☐ facts still correct? ☐ no stronger claim than the source? ☐ citation? ☐

Exercise 3: Read Paper 1's abstract (*individual, 12 minutes, segment 2.3*)

Use the **abstract only** of Paper 1: Boulet N, et al. *Crit Care* 2024;28:429 (in *Papers/*). 8 minutes reading, 4 minutes comparing with your neighbour.

The 15-minute reading order (for later): title (30 s) → abstract (3 min) → figures & tables (4 min) → methods (5 min) → discussion & limitations (2–3 min). Stop early if the paper does not answer your question.

Item	What the abstract says
Design (and blinding)	
P: who was included?	

Item	What the abstract says
I vs C: what was compared?	
Primary outcome (and when)	
Numbers: screened / randomised / analysed	
Main result: estimate with 95% CI and p	
Conclusion: one line	
Does the conclusion match the main result?	☐ yes ☐ no. Why?
Any spin (words stronger than the numbers)?	☐ yes ☐ no. Where?
Registration number	
Does this paper answer our running-example question (28-day mortality)?	☐ yes ☐ no. Why?

Fast finishers: read the **title and abstract** of Paper 2 (Hyun 2023). Which word in the title is stronger than the design allows? Which word does the abstract conclusion use instead?

Spin checklist: words to look for

Data say	Spin says
p = 0.42 for the primary outcome	"a non-significant trend"
primary outcome: no difference	headline = a positive secondary outcome
observational association	"impact", "effect", "reduces"
change within one group only	"patients improved"

Exercise 4: Proposal builder 2 (group, 33 minutes)

Complete Part 1 of the **Research Proposal template**

([Templates/Research_Proposal_Template.docx](#)) and the **literature review worksheet**, building on last week's question, aim and objectives. This sheet is your checklist.

Suggested roles: chair · scribe (types into the template) · searcher (runs your PubMed string on a phone or laptop) · timekeeper.

Time	Task	Done
0–8 min	1. Run your search string (Exercise 1) in PubMed; write the date, the filters and the number of hits	☐
8–16 min	2. Screen titles and abstracts; pick 5 key papers ; export them (<i>Send to</i> → <i>Citation manager</i>) into Zotero	☐
16–22 min	3. Literature matrix: one row per paper (split the papers between members)	☐
22–28 min	4. Problem statement: three short paragraphs (about 300 words): (a) what is known, (b) what is not known (the gap), (c) why it matters <i>here</i> (your local number)	☐

Time	Task	Done
28–31 min	5. Working title and keywords (see below)	☐
31–33 min	One group reads its title and its gap sentence aloud; the room suggests one improvement	☐

Working title and keywords

Working title (10–16 words; population, comparison or exposure, and design visible; no abbreviations; no "impact" or "effect" for observational designs)	
Number of words	
Keywords (3–6; MeSH terms not already in the title)	1. _____ 2. _____ 3. _____ 4. _____

Example (feasible rewrite of the running example): "Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial" (13 words). Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents.

Literature review worksheet: one row per paper

#	Study (first author, year, journal)	Design, size	Compared	Primary outcome: result	Key limitation	Why it matters for our question
1						
2						
3						
4						
5						

Search used for the references (paste the string, the database and the date):

Quality checks before you stop

- [] The problem statement ends with your question; the title names the same population and outcome as the question.
- [] At least one reference is a randomised trial or a systematic review, if one exists; you checked a trial registry for ongoing trials.
- [] References are saved in a reference manager or exported from PubMed (*Send to → Citation manager*). No sentence is copied from a source.

Exit question (*on a sticky note, no names*)

1. Which gives MORE hits: septic shock AND fluids or septic shock OR fluids?
2. What was the primary outcome of Paper 1?

Homework before Session 3 (next Saturday)

- **Group:** finish the five key references and the problem statement, so that Proposal Part 1 is complete with a working title and keywords.
- **Each member:** read one paper with the 15-minute routine and fill its row in the matrix.
- **Everyone:** read the **abstract** of Paper 2 (Hyun 2023, *Ann Intensive Care*) with the Exercise 3 grid.
- Exercise 2 (paraphrasing) if not done in class.
- **Optional:** look up Paper 1's registration (NCT04947904) on ClinicalTrials.gov.